

# Risk-Constrained Next-24-h Empty-Window Recommendations from Occupancy Forecasts: An Offline Building Load-Flexibility Case Study

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**Abstract**—Occupancy forecasts can support building operations only when forecast quality is connected to a constrained operating decision. This paper evaluates next-24-h empty-window recommendations in one office-building time series at 15-min resolution. Historical, tree-based, and sequence models are compared using chronological splits, a 24-h history, and a 24-h forecast horizon. A three-way convex probability blend of a schedule baseline, LightGBM, and a Transformer is selected on validation Empty-class area under the precision-recall curve. Its operating threshold is selected separately on validation to maximize offline safe-load opportunity subject to an interval occupancy-conflict rate no greater than 10% and a minimum one-hour stable empty window. On 43 held-out midnight-issued horizons, the primary blend achieves an all-overlap Empty AUPRC of 0.8514, compared with 0.8497 for the historical baseline and 0.8382 for LightGBM. Its fixed policy recommends 259 intervals, all camera-label-empty, in 14 fully safe labeled windows, with 490.1 kWh of offline safe load opportunity. LightGBM yields a nearly equal 493.9 kWh opportunity but 11 conflict intervals (4.15%). Paired 2,000-replicate daily-block intervals include zero for the primary blend’s AUPRC and opportunity differences versus both baselines. The result is evidence about an offline, validation-selected recommendation policy under a processed, meter-derived HVAC-plus-lighting load proxy. It does not verify energy savings, comfort, physical absence, future safety, or controller performance.

**Index Terms**—occupancy forecasting, building energy management, load flexibility, risk-constrained recommendation, stable empty windows, decision-layer evaluation

## I. Introduction

Building energy-flexibility research considers how operational demand can be adjusted while respecting service constraints [1], [2]. Occupancy is an important contextual signal for that task: it can identify intervals in which a building zone may be empty and, therefore, potentially suitable for a carefully constrained operational action. Occupancy modeling has long been studied for building energy management [3], and recent reviews show a broad forecasting literature spanning data acquisition, estimation, prediction, and evaluation [4]–[6]. Yet a good occupancy forecast alone does not answer the operational question of whether a recommendation can be made at a useful time, for a useful duration, and at an acceptable occupancy-conflict rate.

That distinction matters because the outcome appropriate to a forecasting study is not automatically an energy outcome. Occupancy-prediction-based cooling-

control studies and HVAC-control evaluations can assess a modeled or field operating intervention [7]–[9]. Such evidence is stronger than a retrospective coincidence between a forecast, a camera-derived Empty label, and a processed meter-derived load proxy. This paper deliberately studies the latter: an offline safe load opportunity measure. It sums the HVAC-plus-lighting proxy during intervals that a fixed policy recommends and that are subsequently camera-label-empty. It is neither a counterfactual estimate of energy saved nor a claim that the same load can be shifted without comfort, equipment, or control constraints.

The study asks the following bounded question: for a single office-building time series, how do validation-selected Empty-class forecasts trade off forecast quality, observed occupancy conflict, and offline safe load opportunity when recommendations must form at least one-hour stable empty windows? The evaluation separates two scopes that are often conflated. Forecast discrimination is measured on all overlapping rolling 24-h forecasts. In contrast, an operating policy is selected and evaluated only on non-overlapping midnight-anchored daily horizons, so the policy accounting does not double count repeated predictions for the same interval.

The paper makes three contributions. First, it specifies a leakage-aware, chronological evaluation protocol that maps 24-h Empty probabilities to stable-window recommendations without using controllable loads as predictor inputs. Second, it reports a transparent comparison of a historical schedule baseline, LightGBM, and a validation-selected probability hybrid using both AUPRC and a validation-constrained policy outcome. Third, it exposes the uncertainty and boundary conditions behind the headline result: the primary hybrid has a slightly higher point AUPRC than the two baselines and zero observed conflicts in the held-out policy period, but its paired differences from those baselines are uncertain. The contribution is thus an evaluative case study, not a claim of a new deployed controller or a decision-focused learning algorithm.

Section II positions this evaluation against occupancy forecasting and control research. Sections III and IV define the data, policy, and experimental protocol. Section V reports canonical results, and Sections VI–VII discuss the limits and implications. Appendix A records the status of

the expanded decision-aware and window-aware searches without using them to support the primary conclusion.

## II. Related Work

### A. Occupancy Forecasting

Occupancy forecasting is a mature building-analytics problem rather than a new modeling task. Reviews by Jin et al. and Caballero-Peña et al. organize the literature around forecasting applications, data sources, and evaluation processes [4], [5]. Earlier commercial-building comparisons include stochastic and machine-learning approaches [10], while work based on indoor environmental measurements illustrates the potential and data-quality dependence of sensor-assisted prediction [11]. Sequence models also remain active in the field, including Transformer-based occupancy prediction [12]. These studies motivate a broad baseline set, but they do not establish that one model family is uniformly preferable for every operational rule or building.

The present study accordingly retains a train-only schedule prior alongside nonlinear tabular and sequence models. The tabular reference uses LightGBM [13], the random-forest comparator follows the method of Breiman [14], and the sequence comparator uses a Transformer encoder [15]. A DLinear comparator is also retained because simple linear time-series baselines remain informative [16]. The goal is not to claim algorithmic novelty for any component. Instead, the comparison tests whether combining distinct saved probability forecasts produces an operationally useful, validation-selected policy under the same target and time split.

### B. From Occupancy Prediction to Building Operation

Several studies couple occupancy predictions to a downstream HVAC or ventilation decision. Peng et al. evaluate occupancy-prediction-based cooling control [7]. Esrafilian-Najafabadi and Haghighat analyze how occupancy-model choices relate to HVAC control performance [8], and Zhuang et al. develop probabilistic occupancy forecasts for risk-aware ventilation [17]. Wietzke et al. further connect occupancy prediction to model predictive control and simulation [18]. Those papers demonstrate why a forecast-only score may not summarize a downstream trade-off. They also delimit this paper’s evidence: it does not execute an HVAC action, run a thermal simulation, or measure a counterfactual energy or comfort outcome.

The distinction is especially important when comparing offline studies with operational-system evaluations. Lin et al. report an occupancy-based HVAC control-system evaluation in an office building [9]. This paper has no installation, occupant study, or measured before-and-after intervention. It therefore uses the term opportunity only for a processed meter-derived load proxy coinciding with a recommended and camera-label-empty interval, and it reserves claims about savings or control performance for future intervention or simulation work.

### C. Decision-Layer Evaluation

Decision-focused learning and predict-then-optimize methods train predictive models against a downstream decision objective [19], [20]. The present pipeline does not do this: model parameters are not trained with a differentiable decision loss. It instead fixes candidate probability models, selects blend weights by validation AUPRC, and selects a probability threshold by a validation conflict constraint and offline opportunity objective. The appropriate description is therefore a validation-selected risk-constrained operating policy or decision-layer evaluation. This narrower terminology makes the contribution comparable to risk-aware occupancy work while avoiding a claim that the paper implements decision-focused learning.

## III. Methodology

### A. Data, Target, and Information Set

The source data are the LBNL Building 59 release described by Luo et al. [21] and distributed through Dryad [22]. The repository retains one selected south-zone subset and resamples the streams to 15-min bins. An interval receives the camera-derived occupied label when the maximum selected south-zone camera count is greater than zero. The analysis flips this label so that  $y_{d,t} = 1$  denotes Empty and  $y_{d,t} = 0$  denotes camera-labeled occupied. Thus, Empty means no detected occupant in the selected camera counts, not verified physical absence. The analysis contains 26,413 local-time rows from May 2018 through February 2019. Table I gives the exact split and policy scopes.

Naive source timestamps are interpreted as UTC and converted to America/Los\_Angeles before time features are formed. A solar-radiation peak sanity check supports this interpretation, but the source documentation does not explicitly state the timestamp timezone. Calendar variables are known for future target times. The tabular information set comprises past occupancy/count lags and rolling summaries, issue-time Wi-Fi, temperature, and weather values, and target-time calendar variables. Environmental and association-count sensors are available only up to the forecast issue time and are causally forward-filled. Leading missing values are set to zero. HVAC, lighting, other electrical-load streams, and their aggregate are excluded from predictor features. The retained HVAC and lighting series are used only after forecasting for the offline opportunity calculation.

Each rolling forecast uses 96 historical 15-min intervals and predicts the next 96 intervals. The next-24-h horizon therefore begins 15 min after an issue anchor and ends 24 h after it. It does not imply a 24-h lead time for every recommendation. Chronological train, validation, and test segments are separated by 24.25-h gaps. This prevents a forecast history or target window from straddling a split boundary. It also supports a clean distinction between

TABLE I  
Data and canonical evaluation protocol.

| Item                  | Canonical setting                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Data scope            | Building 59 selected south-zone streams; 26,413 15-min local-time rows, 22 May 2018–21 Feb. 2019 02:00                                        |
| Empty target          | Camera-derived Empty = 1 iff the maximum selected south-zone count is zero in the 15-min bin                                                  |
| Chronological splits  | Train: 23 May–25 Nov. 2018 (32.65% Empty); validation: 28 Nov. 2018–6 Jan. 2019 (29.64%); test: 9 Jan.–21 Feb. 2019 (27.75%)                  |
| Boundary separation   | 24.25-h train-validation and validation-test gaps                                                                                             |
| Forecast scope        | 96 history and 96 horizon steps (24 h each); target offsets 15 min–24 h; all overlapping rolling test forecasts                               |
| Policy scope          | 39 validation and 43 test midnight-anchored, non-overlapping 96-step horizons                                                                 |
| Stable recommendation | Contiguous Empty probability at or above threshold for $\geq 4$ steps (1 h), extracted independently in each policy horizon                   |
| Opportunity proxy     | Causally filled meter-derived $\max(\text{HVAC}_S + \text{lighting}_S, 0) \times 0.25$ h only in recommended-and-camera-label-empty intervals |

model fitting, operating-policy selection, and held-out evaluation.

## B. Forecast Models and Validation-Selected Blend

The comparison includes a train-only historical weekday/time-slot average, LightGBM, random forest, DLinear, and a compact Transformer sequence model. Table II gives the stored implementation settings. Every non-deterministic base model uses seeds 42, 43, and 44; its saved probability is the pointwise mean over those three runs. The saved primary hybrid blends these Empty probabilities, not labels, features, or logits. For time  $t$  it is

$$p_H(t) = 0.15p_{\text{hist}}(t) + 0.60p_{\text{LGBM}}(t) + 0.25p_{\text{Tr}}(t), \quad (1)$$

where  $p_{\text{hist}}$ ,  $p_{\text{LGBM}}$ , and  $p_{\text{Tr}}$  are Empty probabilities from the historical, LightGBM, and Transformer components. The nonnegative weights sum to one. They are the highest-validation-AUPRC point on a declared 0.05 simplex grid of 231 legal weight vectors. Test probabilities are loaded only after this selection. No hybrid component is retrained on validation or test data.

## C. Stable-Window Policy and Offline Opportunity

For a model-specific threshold  $\tau$ , define an initial recommendation  $b_{d,t} = \mathbb{I}[p_{d,t} \geq \tau]$ . Let  $L = 4$  be the number of 15-min intervals in one hour. The policy recommendation  $r_{d,t}$  equals one only when  $b_{d,t}$  belongs to a contiguous run of at least  $L$  intervals within a midnight-anchored daily

horizon. Runs are extracted independently within each 24-h forecast horizon and never join different horizons. The interval occupancy-conflict rate is

$$C = \frac{\sum_{d,t} r_{d,t}(1 - y_{d,t})}{\sum_{d,t} r_{d,t}}, \quad (2)$$

with  $C = 0$  if there are no recommendations.

Let  $\tilde{h}_{d,t}$  and  $\tilde{\ell}_{d,t}$  denote the processed south-HVAC and south-lighting meter streams, respectively, in nominal kW. Define

$$q_{d,t} = 0.25 \max(\tilde{h}_{d,t} + \tilde{\ell}_{d,t}, 0), \quad (3)$$

which is a processed meter-derived 15-min load proxy in kWh. The nominal full-metered accounting convention uses a coefficient of one for each stream, and multiplication by 0.25 h converts kW to interval kWh. Each stream is 8.03% missing before the causal forward fill, so  $q$  includes carried-forward values and is neither a fully observed meter series nor a verified controllable load. The paper reports

$$O_{\text{safe}} = \sum_{d,t} r_{d,t} y_{d,t} q_{d,t}. \quad (4)$$

Equation (4) is an offline overlap measure: it sums the processed load proxy only where a recommendation and a realized camera-derived Empty label coincide. It neither changes a load nor estimates the counterfactual load under a control action. For each model,  $\tau$  is selected from  $\{0.050, 0.075, \dots, 0.950\}$  on validation midnight horizons to maximize  $O_{\text{safe}}$  subject to  $C_{\text{val}} \leq 0.10$ , breaking exact opportunity ties by Empty recall. Because  $O_{\text{safe}}$  requires realized labels and processed loads, it is an offline validation-selection objective, not an online decision-time quantity. The selected threshold is then fixed for all test policy calculations.

## IV. Experimental Design

### A. Evaluation Scopes and Metrics

Forecast metrics use all overlapping rolling test forecasts. The primary metric is Empty-class area under the precision-recall curve (AUPRC), which is appropriate to report alongside other measures for an imbalanced binary class [23]. Empty precision, recall, and F1 in the model-comparison table use a separate  $p_{\text{Empty}} \geq 0.5$  classification cutoff. They do not use the operating-policy threshold in Eq. (2). This distinction prevents a descriptive classifier cutoff from being mistaken for a selected operational rule.

Policy metrics use a different, non-overlapping scope. They use 39 validation and 43 test midnight-anchored 96-step horizons. For every model, the reported threshold is selected only from validation policy horizons. The fixed test policy reports recommended, safe, and conflict intervals, stable recommended and safe windows, coverage, and Eq. (4). The temporal split and fixed-policy evaluation follow the general need to preserve temporal order when estimating time-series performance [24].

TABLE II

Stored model specifications and information availability. All non-deterministic base-model probabilities are pointwise averages over seeds 42, 43, and 44.

| Model                      | Inputs and training set                                                                                                                                 | Stored implementation setting                                                                                                                                                                                  |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Historical average         | Known target weekday and 15-min slot; train partition only                                                                                              | Deterministic mean occupied label by weekday/time slot, with the train mean as fallback                                                                                                                        |
| LightGBM and random forest | Occupancy/count lags and rolling summaries; issue-time sensor values; known-future calendar features; at most 200,000 stratified training rows per seed | LightGBM: 320 trees, learning rate 0.05, 31 leaves, minimum child 50, configured subsample and column-feature fractions 0.85. Random forest: 70 trees, depth 14, minimum leaf 20, square-root feature sampling |
| DLinear                    | Past 96 occupied indicators                                                                                                                             | One linear 96-to-96 mapping; maximum 12 epochs, validation-loss early stopping with patience 4                                                                                                                 |
| Transformer                | Past 96 occupancy/count, issue-time sensor, and calendar values; known-future calendar values                                                           | One Transformer encoder layer, 32-dimensional state, 4 heads, 0.1 dropout; AdamW at $10^{-3}$ ; maximum 12 epochs and validation-loss early stopping with patience 4                                           |

## B. Uncertainty, Diagnostics, and Reproducibility

Uncertainty is quantified with 2,000 paired daily-block bootstrap resamples of the 43 test policy horizons while holding model weights and thresholds fixed. The resulting intervals are conditional on the selected model and policy. They do not account for uncertainty introduced by model selection. Calibration is reported as a held-out diagnostic through Brier score, log loss, and 10-bin quantile expected calibration error. No post-hoc calibrator is fitted. A sensitivity analysis varies the minimum window length from 15 min to 4 h while preserving the validation-selected threshold.

The saved-output path is reproducible without the raw data: it validates aligned prediction exports, repeats the validation-only blend and threshold selection, and regenerates the canonical results. Repository tests check split separation, convex-weight selection, policy accounting, and the prohibition on test-informed candidate selection. The raw Dryad package is not committed, so full end-to-end retraining and independent validation of raw source files are outside this manuscript’s verified scope.

## C. Primary and Exploratory Analyses

The canonical analysis is the validation-selected primary hybrid in Eq. (1) and the same protocol applied to fixed reference models. The repository also contains a test-ranked balanced hybrid, an expanded joint weight-threshold search, and a window-aware search. None of these replaces the canonical primary result. The test-ranked model is excluded from the main analysis, while the joint and window-aware candidates are reported only as status-controlled exploratory material in Appendix A. This separation is necessary because the expanded candidates use an enlarged validation search surface and their present test diagnostics are not fresh, untouched confirmation.

## V. Results

### A. Forecast Quality on Overlapping Test Forecasts

Table III reports the three canonical models central to the paper. The primary hybrid has the highest all-overlap Empty AUPRC point estimate (0.8514), followed closely by the historical average (0.8497). LightGBM is 0.8382. The hybrid also has the lowest reported Brier score among these three models (0.0928). These values summarize 388,032 overlapping forecast rows and should not be interpreted as daily-policy estimates. The extended seven-model comparison is retained in Appendix A for completeness, but the test-ranked balanced hybrid is intentionally omitted from both tables.

The strong schedule baseline is a substantive result. It indicates that weekday and time-of-day structure in this building carries much of the Empty-class predictive signal. The hybrid’s point AUPRC margins are small: 0.0017 over the historical baseline and 0.0132 over LightGBM. Section V-C therefore evaluates these differences at the daily-horizon level rather than treating their ranking in Table III as decisive evidence of superiority.

### B. Validation-Selected Policy Outcomes

Table IV and Fig. 1 report the policy scope: 43 non-overlapping test days and 4,128 15-min intervals. Every listed threshold met the validation 10% interval-conflict constraint. The primary hybrid used  $\tau = 0.875$  after a validation conflict rate of 8.75%. On test, it recommended 259 intervals. All 259 were camera-label-empty. Those intervals formed 14 recommended windows, all fully safe under the camera-derived label, and produced 490.1 kWh of offline safe load opportunity (11.40 kWh/day).

LightGBM produced a nearly equal 493.9 kWh opportunity, but 11 of 265 recommended intervals were camera-labeled occupied (4.15%) and three of 19 recommended windows contained an occupancy conflict. The historical average was equally conflict-free under this label in the test period but recommended much less processed load-proxy opportunity (94.6 kWh). Figure 1 is deliberately restricted

TABLE III

Canonical three-model forecast comparison on all overlapping held-out test forecasts (Empty = 1). Precision, recall, and F1 use a descriptive 0.5 probability cutoff. The policy thresholds in Table IV are selected separately on validation.

| Model              | Empty AUPRC | Empty Brier | Empty precision | Empty recall | Empty F1 |
|--------------------|-------------|-------------|-----------------|--------------|----------|
| Historical average | 0.8497      | 0.0974      | 0.7264          | 0.8015       | 0.7621   |
| LightGBM           | 0.8382      | 0.0956      | 0.7610          | 0.7593       | 0.7601   |
| Primary hybrid     | 0.8514      | 0.0928      | 0.7677          | 0.7556       | 0.7616   |

TABLE IV

Validation-selected one-hour stable-window policies evaluated on 43 non-overlapping test days. Every threshold maximizes validation safe opportunity subject to validation interval conflict  $\leq 10\%$ . “Safe” means recommended and camera-label-empty. Opportunity is an offline processed-load-proxy overlap, not measured energy savings.

| Model              | $\tau$ | Val. conflict | Test conflict  | Safe opportunity | Safe/rec. intervals | Safe/rec. windows | kWh/day |
|--------------------|--------|---------------|----------------|------------------|---------------------|-------------------|---------|
| Historical average | 0.950  | 8.33%         | 0.00% (0/66)   | 94.6             | 66/66               | 12/12             | 2.20    |
| LightGBM           | 0.950  | 9.61%         | 4.15% (11/265) | 493.9            | 254/265             | 16/19             | 11.49   |
| Primary hybrid     | 0.875  | 8.75%         | 0.00% (0/259)  | 490.1            | 259/259             | 14/14             | 11.40   |

to these three pre-specified models; it does not display test-ranked or retrospective exploratory candidates.

### C. Conditional Uncertainty and Diagnostics

The daily-block uncertainty results in Table V condition on the previously selected weights and thresholds. The primary hybrid’s daily-horizon AUPRC is 0.8522 with a 95% percentile interval of [0.7918, 0.9016]. This differs from the all-overlap value in Table III because the aggregation unit differs. The paired AUPRC intervals for primary minus LightGBM and primary minus historical average both cross zero. The paired safe-opportunity interval also crosses zero for primary minus LightGBM, although the primary-minus-historical comparison is positive over its reported interval. Figure 2 makes the contrast explicit rather than visually magnifying the small AUPRC point differences.

The primary hybrid’s held-out calibration diagnostics are Brier score 0.0928, log loss 0.2984, and 10-bin quantile ECE 0.0253. No recalibration was applied. At the fixed primary threshold, a one-hour or two-hour minimum window retains 490.1 kWh and zero observed camera-label conflicts, while a four-hour requirement retains 407.8 kWh and 11 fully safe windows. These are sensitivity results at a fixed threshold, not re-optimized policies. Finally, every observed primary-hybrid camera-label conflict block is zero, so a nonparametric bootstrap also returns a zero conflict interval. Under an unverified independent-trial assumption, zero events yield one-sided 95% upper bounds of 1.15% for 259 intervals and 19.3% for 14 windows. Intervals and windows are clustered, so neither bound is a deployment-safety guarantee.

## VI. Discussion

### A. What the Comparison Does and Does Not Show

The historical average’s performance is informative rather than an inconvenience to be optimized away. A stable schedule can produce a highly competitive Empty

forecast in a single office building, and the small point AUPRC advantage of the primary hybrid is not decisive under the reported paired daily-block intervals. The hybrid’s practical distinction in this period is not a large forecast score gain: it selects an operating point with approximately the same processed load-proxy opportunity as LightGBM and no camera-labeled occupied recommendations. This is a reason to report forecast and policy metrics together, not a reason to declare a universally superior model.

The offline opportunity formulation also provides a useful guardrail for interpretation. It makes the recommendation rule, duration requirement, conflict denominator, and load proxy inspectable. In particular, it avoids calling every correctly predicted Empty interval operationally useful. A short isolated interval is removed by the stable-window rule. At the same time, Eq. (4) has no counterfactual. The processed HVAC and lighting values may not be controllable at that moment, shifting a load may create a later penalty, and a real intervention would require constraints on equipment, indoor conditions, tariffs, and occupants. This paper therefore does not estimate savings, flexibility capacity, carbon reduction, or comfort preservation.

### B. Validity Limits

Several limitations constrain the scope of the result. First, the data cover selected south zones in one building and a 43-day test policy period. External and cross-building validation are unavailable. Second, source timestamps are interpreted as UTC before Pacific conversion on the basis of a solar-pattern diagnostic, but the data documentation does not explicitly confirm that timezone. Third, the camera-count label can miss occupants and does not establish physical absence. Results called safe or conflict-free apply only to that label. Fourth, the Wi-Fi association stream has substantial missingness before causal filling, and the HVAC and lighting streams used

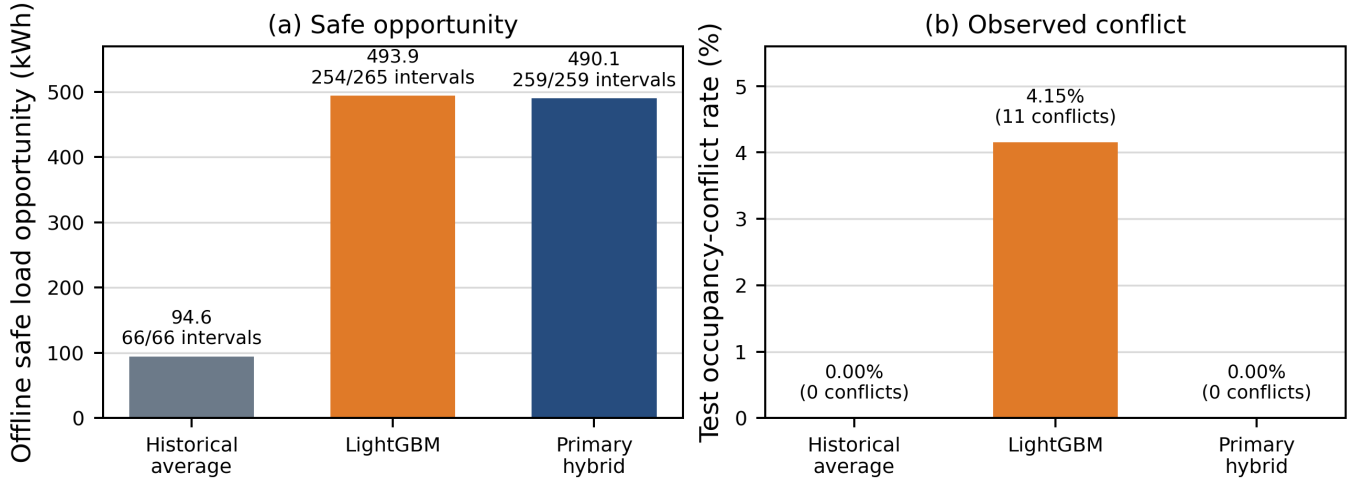


Fig. 1. Held-out policy outcomes for the schedule baseline, LightGBM, and the validation-selected primary hybrid. Thresholds were selected on 39 validation daily horizons to maximize offline safe load opportunity subject to an interval conflict rate no greater than 10%; the plotted values are one fixed evaluation on 43 test horizons. “Safe” means recommended and subsequently camera-label-empty, not physical absence, a verified load shift, or energy saving.

TABLE V

Conditional daily-block uncertainty from 2,000 paired bootstrap resamples of 43 non-overlapping test days. Daily AUPRC is not the all-overlap AUPRC in Table III. All intervals condition on selected weights and thresholds.

| Estimate type   | Quantity                               | Point estimate | 95% low | 95% high |
|-----------------|----------------------------------------|----------------|---------|----------|
| Primary model   | Daily-horizon Empty AUPRC              | 0.8522         | 0.7918  | 0.9016   |
| Primary model   | Safe opportunity (kWh)                 | 490.1          | 153.1   | 966.7    |
| Paired contrast | Primary – LightGBM AUPRC               | +0.0163        | -0.0050 | +0.0426  |
| Paired contrast | Primary – historical AUPRC             | +0.0042        | -0.0304 | +0.0377  |
| Paired contrast | Primary – LightGBM opportunity (kWh)   | -3.8           | -208.2  | +174.6   |
| Paired contrast | Primary – historical opportunity (kWh) | +395.5         | +127.0  | +786.2   |

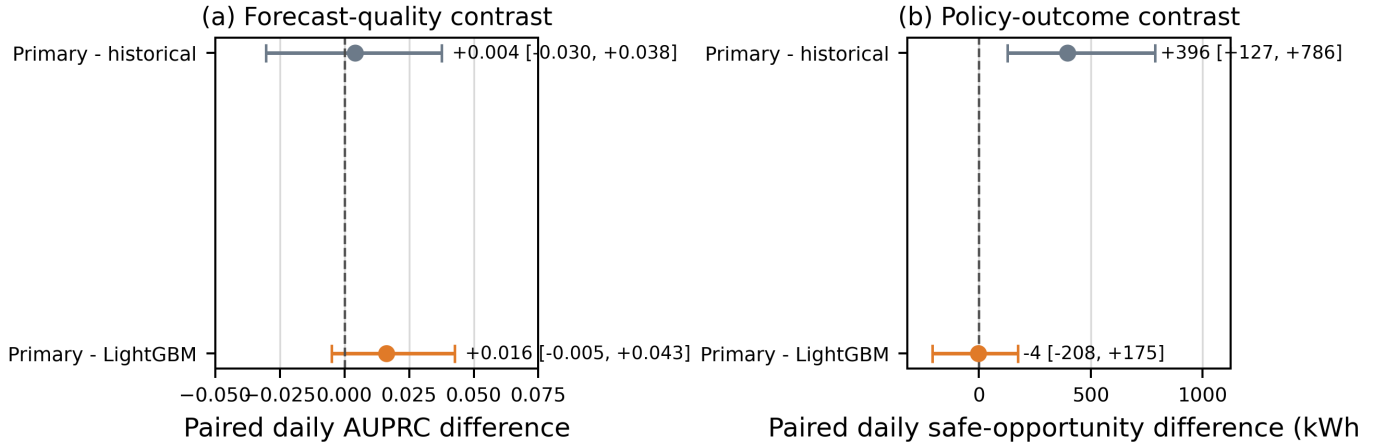


Fig. 2. Paired 95% daily-block bootstrap intervals (2,000 resamples) for the primary hybrid minus each canonical baseline. The dashed line denotes no difference. Intervals condition on fixed, validation-selected model weights and operating thresholds; they do not include model-selection uncertainty.

in  $q$  are each 8.03% missing before the same causal fill. The study does not establish robustness to different sensor availability or load-imputation patterns. Fifth, the raw source package is absent from the repository, so

the saved-output workflow is deterministic but does not independently verify raw-data retraining.

The statistical limitations are also material. The bootstrap conditions on selected weights and thresholds and

cannot generate unseen conflicts when all resampled blocks have zero conflicts. Hybrid-specific seed dispersion and full hybrid rolling-origin reconstruction cannot be recovered from the saved exports because aligned per-seed and rolling Transformer predictions are missing. In addition, the primary weights were reconstructed as the top point on the declared validation grid after an earlier staged script had hard-coded the same values. The available artifacts verify the reconstructed validation-before-test procedure, but the study is not a preregistered, fully blinded confirmatory evaluation with a fresh test firewall.

### C. Status of Expanded Searches and Next Evidence

The repository’s decision-aware joint search and window-aware search are not evidence for replacing the canonical policy. Both apply selection rules to validation outputs, but they enlarge the search surface, and their current test diagnostics are no longer a fresh untouched evaluation. Appendix A labels their point estimates as exploratory or as frozen future challengers rather than performance improvements. This distinction is essential because decision-focused learning would require downstream-objective training, which this pipeline does not perform [19], [20].

The next defensible experiment is a prospectively locked evaluation: freeze the candidate, threshold, stable-window rule, and load-accounting definition. Test them on a later untouched chronological period or a new building. Couple the recommendation to a simulator or intervention with explicit comfort and equipment constraints. That design would make it possible to evaluate the control claims that prior occupancy-aware HVAC and ventilation studies are designed to investigate [7], [17], [18].

## VII. Conclusion

This paper presents a transparent, risk-constrained evaluation from next-24-h occupancy forecasts to stable Empty-window recommendations. In one Building 59 case study, a validation-selected historical-LightGBM-Transformer probability blend achieved 0.8514 all-overlap Empty AUPRC and, at its validation-selected one-hour policy, 490.1 kWh of offline safe load opportunity with zero camera-label conflicts across 259 recommended test intervals. The historical schedule baseline remained nearly equal in AUPRC, and the primary hybrid’s paired differences from LightGBM and the schedule baseline were not decisive for forecast quality or for opportunity versus LightGBM. The contribution is therefore a reproducible forecast-to-policy assessment with explicit evidence boundaries. It does not demonstrate energy savings, comfort preservation, physical absence, controller safety, or generalization. Before submission as an operational method, the exact policy should be frozen and evaluated prospectively on an untouched period or additional buildings with a counterfactual control, simulation, or field-intervention outcome.

## Appendix A

### Extended Results and Exploratory-Analysis Status

Table VI preserves the complete canonical comparison set. The test-ranked balanced hybrid is not shown because its architecture was selected after inspection of test ranking and is supplementary by definition. The primary hybrid remains the validation-selected canonical model in the manuscript.

The repository additionally contains a joint weight-threshold search over 231 convex blend weights and 37 probability thresholds (8,547 validation pairs) and a window-aware search over the resulting validation surface. These analyses have useful engineering value because their constraints are applied to validation data and the code rejects test-informed selection inputs. They nevertheless enlarge the search surface and were evaluated against a test period that is no longer a fresh independent confirmation. The decision-optimal current-test point estimates (623.5 and 650.4 kWh for two joint-search variants) and the window-aware challenger’s current-test point estimate (511.6 kWh with 2.49% interval conflict) are therefore retrospective diagnostics, not headline efficacy results.

a) Status controls.: The test-ranked balanced hybrid is supplementary only because its architecture family was selected after inspection of test ranking. The joint search is exploratory and cannot replace the canonical primary policy. The window-aware candidate is a frozen future challenger, not a confirmed result. None of these searches trains a downstream decision loss. A future evaluation may use a frozen candidate and its declared constraints on a later untouched chronological period, ideally with control, comfort, and counterfactual energy measurements.

### Data and Code Availability

The underlying Building 59 data package is available from Dryad [22]. Raw data are not committed to this repository. The submitted artifact should include the repository revision, the canonical saved prediction and result exports, and the scripts used to regenerate the manuscript figures and tables. A persistent archival URL must be inserted here before submission.

### Author Contributions, Funding, and Competing Interests

To preserve anonymous review, contributor identities, funding information, and conflict-of-interest declarations are omitted from this draft. The corresponding author must replace this statement with the target venue’s required declarations before submission.

## References

- [1] Y. Chen, P. Xu, J. Gu, F. Schmidt, and W. Li, “Measures to improve energy demand flexibility in buildings for demand response (DR): A review,” *Energy and Buildings*, vol. 177, pp. 125–139, 2018.
- [2] R. G. Junker, A. G. Azar, R. A. Lopes, K. B. Lindberg, G. Reynders, R. Relan, and H. Madsen, “Characterizing the energy flexibility of buildings and districts,” *Applied Energy*, vol. 225, pp. 175–182, 2018.

TABLE VI

Extended canonical model comparison on all overlapping held-out test forecasts (Empty = 1). The test-ranked balanced hybrid is excluded because it is supplementary only.

| Model                      | Empty AUPRC | Empty precision | Empty recall | Empty F1 |
|----------------------------|-------------|-----------------|--------------|----------|
| Historical average         | 0.8497      | 0.7264          | 0.8015       | 0.7621   |
| LightGBM                   | 0.8382      | 0.7610          | 0.7593       | 0.7601   |
| Random forest              | 0.8329      | 0.7510          | 0.8019       | 0.7756   |
| Original Transformer       | 0.7621      | 0.7253          | 0.6022       | 0.6581   |
| DLinear                    | 0.5933      | 0.5764          | 0.6657       | 0.6179   |
| Seasonal-Transformer blend | 0.8490      | 0.7465          | 0.7408       | 0.7436   |
| Primary hybrid             | 0.8514      | 0.7677          | 0.7556       | 0.7616   |

- [3] V. L. Erickson, M. Á. Carreira-Perpiñán, and A. E. Cerpa, "Occupancy modeling and prediction for building energy management," *ACM Transactions on Sensor Networks*, vol. 10, no. 3, 2014, art. no. 42, 28 pp.
- [4] Y. Jin, D. Yan, A. Chong, B. Dong, and J. An, "Building occupancy forecasting: A systematical and critical review," *Energy and Buildings*, vol. 251, p. 111345, 2021.
- [5] J. Caballero-Peña, G. Osma-Pinto, J. M. Rey, S. Nagarsheth, N. Henao, and K. Agbossou, "Analysis of the building occupancy estimation and prediction process: A systematic review," *Energy and Buildings*, vol. 313, p. 114230, 2024.
- [6] T. Li, X. Liu, G. Li, X. Wang, J. Ma, C. Xu, and Q. Mao, "A systematic review and comprehensive analysis of building occupancy prediction," *Renewable and Sustainable Energy Reviews*, vol. 193, p. 114284, 2024.
- [7] Y. Peng, A. Rysanek, Z. Nagy, and A. Schlüter, "Using machine learning techniques for occupancy-prediction-based cooling control in office buildings," *Applied Energy*, vol. 211, pp. 1343–1358, 2018.
- [8] M. Esrafilian-Najafabadi and F. Haghighat, "Impact of occupancy prediction models on building HVAC control system performance: Application of machine learning techniques," *Energy and Buildings*, vol. 257, p. 111808, 2022.
- [9] G. Lin, A. Casillas, M. Sheng, and J. Granderson, "Performance evaluation of an occupancy-based HVAC control system in an office building," *Energies*, vol. 16, no. 20, p. 7088, 2023.
- [10] Z. Li and B. Dong, "Short term predictions of occupancy in commercial buildings—performance analysis for stochastic models and machine learning approaches," *Energy and Buildings*, vol. 158, pp. 268–281, 2018.
- [11] S. H. Ryu and H. J. Moon, "Development of an occupancy prediction model using indoor environmental data based on machine learning techniques," *Building and Environment*, vol. 107, pp. 1–9, 2016.
- [12] K. Sun, I. Qaisar, M. A. Khan, T. Xing, and Q. Zhao, "Building occupancy number prediction: A transformer approach," *Building and Environment*, vol. 244, p. 110807, 2023.
- [13] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154.
- [14] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.
- [16] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, pp. 11 121–11 128, 2023.
- [17] C. Zhuang, R. Choudhary, and A. Mavrogianni, "Probabilistic occupancy forecasting for risk-aware optimal ventilation through autoencoder bayesian deep neural networks," *Building and Environment*, vol. 219, p. 109207, 2022.
- [18] T. Wietzke, J. Gall, and K. Graichen, "Occupancy prediction for building energy systems with latent force models," *Energy and Buildings*, vol. 307, p. 113968, 2024.
- [19] B. Wilder, B. Dilkina, and M. Tambe, "Melding the data-decisions pipeline: Decision-focused learning for combinatorial optimization," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 1, pp. 1658–1665, 2019.
- [20] A. N. Elmachtoub and P. Grigas, "Smart "predict, then optimize"," *Management Science*, vol. 68, no. 1, pp. 9–26, 2022.
- [21] N. Luo, Z. Wang, D. Blum, C. Weyandt, N. Bourassa, M. A. Piette, and T. Hong, "A three-year dataset supporting research on building energy management and occupancy analytics," *Scientific Data*, vol. 9, no. 1, p. 156, 2022.
- [22] T. Hong, N. Luo, D. Blum, and Z. Wang, "A three-year building operational performance dataset for informing energy efficiency," *Dryad Dataset*, ver. 6, 2022. [Online]. Available: <https://doi.org/10.7941/D1N33Q>
- [23] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.
- [24] V. Cerqueira, L. Torgo, and I. Mozetič, "Evaluating time series forecasting models: An empirical study on performance estimation methods," *Machine Learning*, vol. 109, no. 11, pp. 1997–2028, 2020.